Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Agentic AI · LangGraph · Multi-Agent · RAG · Evaluation Harnesses · Guardrails · MCP · Enterprise SaaS · 9yr Production

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PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience designing, building, and shipping autonomous and semi-autonomous AI agents, multi-agent orchestration systems, RAG infrastructure, and evaluation harnesses - exactly the work this role describes. Leasel is the multi-agent proof point: an 8-agent LangGraph StateGraph (planning, reasoning, memory, tool invocation) with real conditional routing, dynamic task delegation based on confidence, shared long-term memory across agents and sessions, and deterministic HITL guardrails that prevent unsafe tool calls and hallucination-driven decisions on regulated paths - the ReAct and planner-executor patterns Salesforce's agentic systems require at enterprise scale. AgentProbe is the evaluation harness: a production platform with a 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-Judge scoring, pass@k multi-trial harness, CI-integrated regression gate blocking drift before deployment, and MCP integration for AI-to-system data retrieval - purpose-built for non-deterministic systems where evaluation is a first-class engineering practice. At Entrata, I owned production enterprise SaaS systems for 9 years (AWS, Docker microservices, RabbitMQ event-driven async, FastAPI) at 99.5% uptime - the distributed architecture and production reliability standards enterprise-grade agentic systems require. I mentor engineers on agent development patterns, lead code reviews, and own complex features from design through incident response and RCA. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Agentic AI & Orchestration: Python (9yr), LangGraph, LangChain, Multi-Agent Frameworks, Agent Orchestration, Planning & Reasoning, Long-Term Memory, Tool Invocation, Tool-Calling Patterns, ReAct, Planner-Executor, Conditional Routing, HITL, Guardrails, MCP (Model Context Protocol), Anthropic API, OpenAI API, Ollama, Prompt Engineering, Context Engineering, Claude Code

Evaluation & LLMOps: Evaluation Harnesses, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Groundedness, pass@k Harness, CI Regression Gate, Drift Detection, Non-Deterministic Testing, RAG Pipelines, ChromaDB, Vector Databases, Embeddings, Chunking Strategies, Semantic Search, NLP, BERT, spaCy, PyTorch, scikit-learn

Enterprise Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, FastAPI, REST APIs, RabbitMQ (async, event-driven), Redis, PostgreSQL, MySQL, Oracle, GitHub Actions CI/CD, Distributed Systems, Performance Engineering, Incident Response & RCA, PHI/HIPAA Compliance, TypeScript, React.js, Linux, Git, Agile/Scrum

PROJECTS

Leasel: Multi-Agent System - Planning, Memory, Tool Use & Guardrails

Multi-Agent · ReAct · Long-Term Memory · HITL · Production

Python · LangGraph (8-agent) · FastAPI · TF-IDF RAG · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production multi-agent system demonstrating the full agentic AI stack - planning, reasoning, memory, tool invocation, guardrails, and human-in-the-loop safety at enterprise grade. **Agent orchestration:** 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) with real conditional routing and dynamic task delegation based on confidence scoring - the planner-executor pattern. **Long-term memory:** shared SQLite knowledge repository across all agents - documents, settings, incidents, and audit logs persisted across sessions. **Tool invocation:** agents invoke retrieval, structured data, and configuration tools with grounded, citation-backed outputs. **Guardrails:** deterministic HITL escalation prevents unsafe or hallucinated tool calls on regulated decision paths. 93.6% precision, 87.9% F1 across 1,240 test cases.

AgentProbe: Production Evaluation Harness - Non-Deterministic AI Systems

Evaluation Harness · LLMOps · CI Gate · MCP · Open Source

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production evaluation platform for non-deterministic AI systems - built specifically to make evaluation a first-class engineering practice. **Evaluation harness:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring; pass@k multi-trial harness with multi-sampling to surface variance in non-deterministic outputs. **CI regression gate:** GitHub Actions integration blocks quality drift and unsafe output patterns before deployment. **RAG infrastructure:** ChromaDB vector database, chunking strategy optimization, embedding selection, semantic search. **MCP integration:** unified AI-to-system data retrieval across providers. Multi-provider LLM routing (Anthropic, OpenAI, Ollama). Zero regressions across 1,240+ test cases. Open-source, enterprise-deployable.

AI on FHIR: Enterprise API Integration - Agentic Tool Use & HIPAA

Enterprise APIs · Tool Use · HIPAA · Full Stack

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack agentic system integrating AI with enterprise REST APIs - LLM-powered tool invocation translates natural-language queries into structured FHIR API calls; HIPAA-compliant RBAC with authentication and full data isolation; ~60% reduction in manual retrieval time; Docker containerization. Demonstrates the API integration and secure tool-use pattern (agents safely reading and writing data in enterprise systems) Salesforce's Agentforce platform implements at scale.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production Agentic AI Delivery

- Designed and deployed a production agentic AI system (LangGraph, FastAPI, Python) for an external customer - end-to-end ownership from requirements through architecture, implementation, evaluation harness, and production deployment in weeks as sole technical owner. Built guardrails, validation gates, and human-review escalation for regulated decision paths; contributed documentation and runbooks for ongoing operation.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and software engineering; delivered complex technical outputs for non-technical stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Enterprise SaaS production systems** - AWS S3/EC2/RDS/Glue; Docker microservices (50% scalability improvement, 30% deployment time reduction); RabbitMQ event-driven async with retry logic and idempotency guarantees; Redis caching; 99.5% uptime across multi-tenant SaaS; 100% PHI/HIPAA compliance; zero security incidents across 7yr.
- **Incident response, RCA, and production reliability** - owned production incident response for HUD compliance modules; built internal debugging tools reducing triage time 40%; maintained bug count below 10 for 6+ consecutive months; 200+ SQL optimizations (30% performance gain); GitHub Actions CI/CD with automated testing throughout; Python/Bash automation eliminating 80% of manual workload.
- **Technical coaching and cross-functional delivery** - mentored 5+ engineers on system design, code quality, and production practices; 300+ peer code reviews; 120+ Agile sprints with product, compliance, and operations stakeholders; 20+ full-stack production modules shipped end-to-end; leveraged Claude Code to accelerate engineering delivery.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017